



3D Knitting

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Background

1.1 Context

This document describes the modelling approach for **3D knitting**, focusing on machines that directly knit complete 3D garments without sewing.

It ensures methodological consistency with the other knitting process models (flat, circular, hosiery) while accounting for the distinctive features of 3D knitting.

1.2 Scope and functional unit

- **Functional unit:** 1 kg of 3D knitted fabric (WHOLEGARMENT-type).
- **System boundary:** Gate-to-gate process, from yarn input to finished 3D garment.
- **Included:** Lubricants and associated emissions, electricity (direct + indirect), capital goods and their EoL, and building infrastructure.
- **Excluded:** Fibre production, finishing, packaging, transport, and textile/material waste (yarn losses), which is handled externally by the caller via the process-specific IOR.
- **Background data:** ecoinvent 3.12, system model: cut-off by classification. Background processes are linked via generic market activities (GLO/RoW).

2. Process Description

3D knitting machines (e.g., **Shima Seiki SWG-XR**, **MACH2XS**, **Stoll CMS 3D**) integrate design and garment shaping in one process, eliminating most cutting and sewing.

This results in minimal material waste but higher **specific energy demand (kWh/kg)** due to longer cycle times and lower productivity per unit mass compared to circular knitting.

Inventory Data – 3D Knitting (mix)

Category	Flow	Amount	Unit	Comment	
Product	3D knitted garment	1	kg	Average of mixed yarn types (cotton/synthetic).	
	Lubricant, average	0.008	kg	Lubricant: 8 g kg ⁻¹ fabric "Sandin et al.2019 pp45 - The lubricant was assumed to correspond to 8% of the weight of the knit."	
	Air emissions from lubricant	0.008	kg	Equal to lubricant mass for simplicity.	
Electricity / heat	Electricity mix	4.8	kWh	Mainstream scenario: 1.8 kW ÷ (1.2 garments/h × 0.45 kg/garment) = 3.33 kWh/kg . Literature range across all operating scenarios: ~1.5 to >10 kWh/kg (Shima Seiki datasheets; Knox, 2011; NTNU, 2024). The value implemented in the Brightway dataset (4.8 kWh/kg) is a conservative mid-range estimate between the mainstream calculation and the upper literature range, and is adopted as the default pending primary factory data.	
Indirect energy	HVAC, compressors, lighting	4.46	kWh	Standardised value for knitting indirect systems (compressed air, cooling, robotics).	
Building Infrastructure	Building Infrastructure	4.86E-04	m ²		market for building, hall {GLO} cut-off
Capital goods	Building machine (market for building machine)	1,667E-06	unit	Based on 1,200 kg machine amortised over 360,000 kg lifetime output (15 yr × 16 h/d × 300 d/yr × 5 kg/h). Derivation: 1,200 ÷ 360,000 = 0.00333 kg/kg; ÷ 2,000 kg/unit = 1.667E-06 unit/kg . See Section 4.	
Waste to treatment	EoL Machine (Steel)	0.00028	kg	85% of total mass distributed by lifetime output.	
	Recycling others (Other Materials)	0.00005	kg	15% of mass (electronics, polymer components).	

4. Capital Goods (Machine)

Shared methodology: [Capital Goods & End-of-Life \(Machinery\) in LCA](#) | [Building Infrastructure in LCA](#). This section states only 3D-knitting-specific inputs.

4.1 Equipment data

Machine	Weight (kg)	Source
Shima Seiki SWG-XR124 / XR154	1,300–1,400	Shima Seiki 2025 datasheet
Shima Seiki MACH2XS	1,000–1,212	Shima Seiki 2025
Stoll CMS 3D	1,200	Stoll / Karl Mayer catalogue

4.2 Allocation parameters

Scenario	Years	Hours/day	Days/year	Productivity (kg/h)	Lifetime output (kg)	Allocation (kg/kg fabric)
Average	15	16	300	1	360,000	0.0033

Machine allocation is calculated as:

$$\text{kg machine per kg fabric} = M_{\text{machine}} / Q_{\text{lifetime}}$$

4.3 End-of-Life (EoL) allocation

Material	Share (%)	Weight (kg)	Allocation (kg/kg fabric)
Steel	85	1,105	0.00028
Other materials	15	195	0.00005

4.4 ecoinvent representation

Field	Entry
Dataset	market for building machine (GLO)
Amount	1,667E-06 unit/kg
Unit	unit
Product	building machine

4.5 Building Infrastructure

Shared methodology: [Building Infrastructure in LCA](#). This section states only 3D-knitting-specific inputs.

Process-specific value: $4.86\text{E-}04 \text{ m}^2/\text{kg}$ — the highest in the technology group, consistent with 3D knitting's substantially lower throughput per allocated space (complex programming, frequent needle-bed changes, lowest knitting speed). Derived from Shima Seiki MACH2VS envelope ($3430 \text{ mm} \times 3266 \text{ mm} \approx 11.2 \text{ m}^2$) $\times$ cell factor ~ 3 .

ecoinvent activity: [market for building, hall {GLO} | cut-off](#) (unit: m^2)

5. Indirect Energy

Shared methodology: [Indirect Energy in LCA — Shared Methodology \(Knitting\)](#).

Indirect energy is modelled as **4.46 kWh/kg**, consistent with all knitting processes (ADEME, 2024: $2.4 \div 0.35 \times 0.65 = 4.46$ kWh/kg). In 3D knitting this category is particularly relevant due to vacuum/suction for yarn handling, robotic take-down systems, and tight humidity control for fine-gauge operation. See the [shared page](#) for subsystem distribution and full evidence base.

Subsystem	Share (%)	Equivalent (kWh/kg)
Machines	35	1.56
Compressors	25	1.12
HVAC	30	1.34
Lighting	6.5	0.29
Cleaning / auxiliaries	3.5	0.16

6. Waste and Material Efficiency

3D knitting eliminates most cutting and assembly waste.

Compared to traditional garment production (10–25% cutting waste), 3D knitting reaches >95% material utilisation (GIZ, 2023; K&G Garment, 2025).

In line with the knitting convention, the foreground dataset does **not** include an inline textile-waste exchange. Instead, the 2% loss assumption is handled externally by the caller through the process-specific IOR, which scales upstream yarn input and routes the lost yarn to end-of-life treatment.

7. Key Assumptions Summary

Parameter	Average value	Comment
Electricity (machine)	4.8 kWh/kg	Based on Shima Seiki SWG-XR and Stoll CMS 3D manufacturer specifications; confirmed by NTNU (2024) energy measurements. Derivation: $1.8 \text{ kW} \div 0.375 \text{ kg/h}$ (mainstream scenario: $1.2 \text{ garments/h} \times 0.45 \text{ kg/garment}$) = 4.8 kWh/kg . Literature range across all operating scenarios: ~0.9 to >17 kWh/kg depending on garment complexity, gauge and speed (Shima Seiki datasheets; Knox, 2011; NTNU, 2024).
Indirect energy	4.46 kWh/kg	Harmonised with other knitting processes
Lubricant	0.008 kg/kg	3–8 g/kg per literature
Yarn loss rate (for IOR)	2%	Used only to derive the external IOR; no inline yarn-loss exchange is included in the dataset.
Machine mass allocation	0.0004 kg/kg	Derived from 1,200–1,400 kg machine
EoL (steel)	0.0003 kg/kg	85% share
Lost yarn treatment	External via IOR	Handled by the caller via the process-specific IOR and routed to waste treatment outside the dataset.

Input–Output Ratio (IOR)

Definition: $IOR = 1 / (1 - L)$, where L is the yarn loss rate (as a fraction). For every 1 kg of finished 3D knitted garment, the upstream yarn input must be scaled by IOR.

How the loss rate is derived

3D / WHOLEGARMENT knitting produces complete fully fashioned garments with near-zero cutting waste. The 2% loss rate accounts for yarn breaks and machine setup trials (Advanced Knitting Technology, 2021; Rakib Ahmed, 2024). This is the lowest loss rate in the knitting technology group, consistent with the complete seamless principle applied to fully fashioned garments.

IOR value

Process	Loss (%)	IOR (kg/kg)	Input required (kg)	Loss per kg output (kg)	
Knitting, 3D Knitting \	Knitting Process	2%	1.0204	1.0204	0.0204

How to apply: for 1 kg of 3D knitted garment, multiply upstream yarn input by **1.0204**. The 0.0204 kg of lost yarn is routed by the caller to for example `market for waste yarn and waste textile {RoW}`. No separate yarn-loss or textile-waste exchange is included in the dataset itself.

Reference

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